

Dashboarding

Knowledge objectives

1. Explain the need and benefits of data visualization
2. Define dashboard
3. Explain five desirable characteristics of a dashboard
4. Define KPI
5. Explain the SMART characteristics of a KPI
6. Explain seven problems in a chart
7. Explain ten elements in a good chart
8. Explain eleven design principles of graphs

Understanding Objectives

1. Distinguish a good from a bad dashboard design
2. Identify a misleading chart

DEFINITION AND CHARACTERISTICS

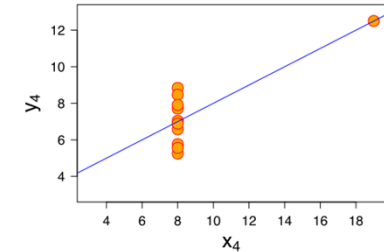
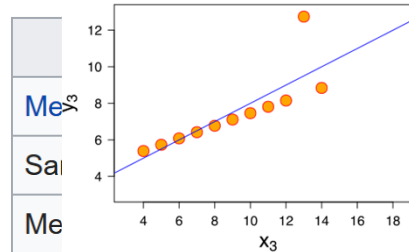
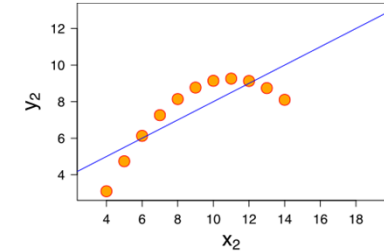
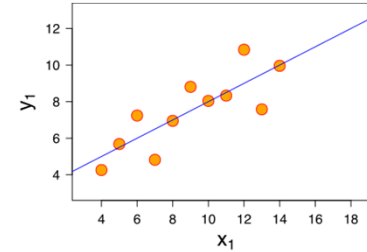
Visualization

“The **use of computers** or techniques for **comprehending data** or to extract knowledge from the results of **simulations, computations, or measurements.**”

McCormick et al.

Need

I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89



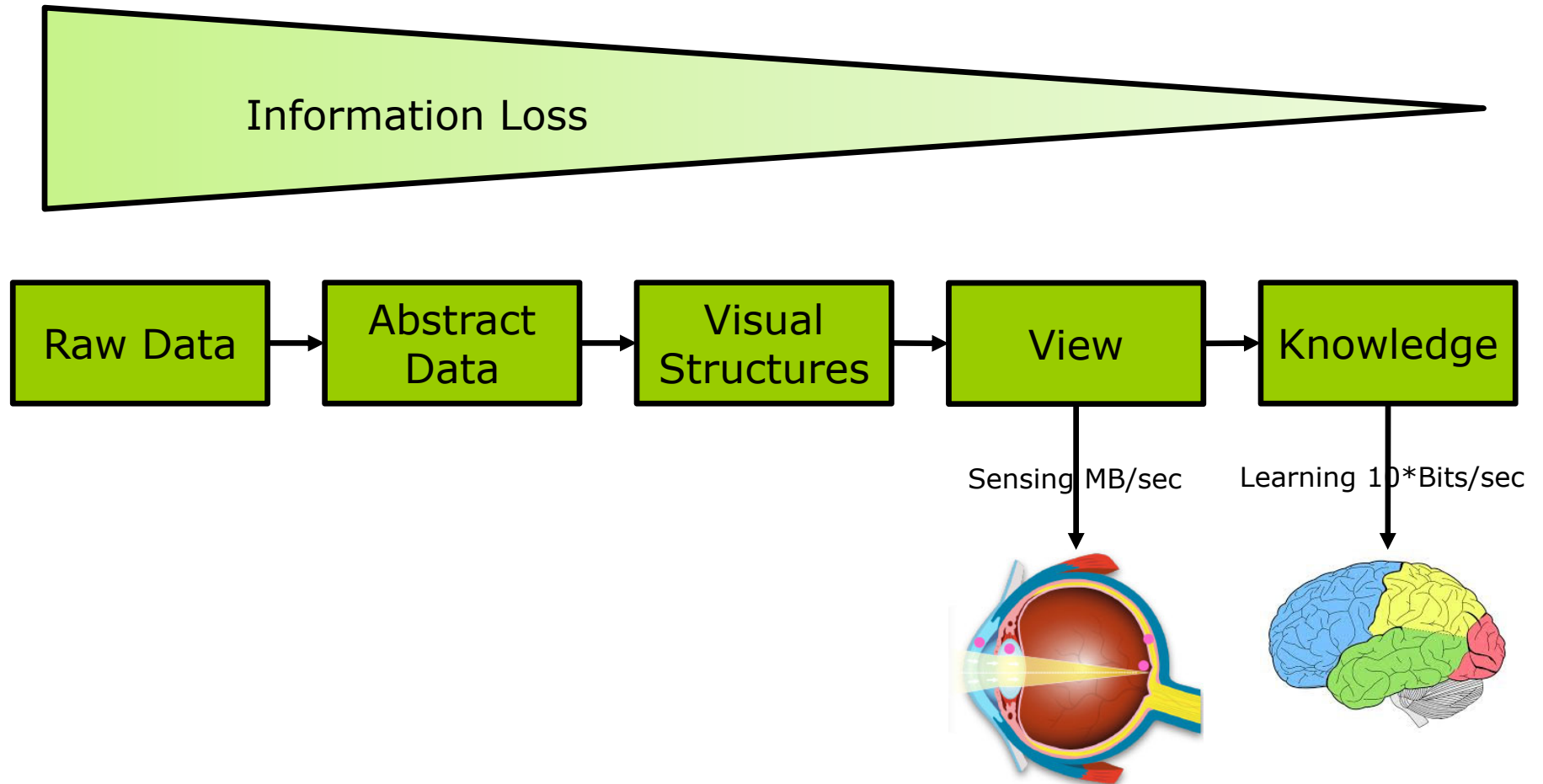
Sample variance of y	4.125
Correlation between x and y	0.816
Linear regression line	$y = 3.00 + 0.500x$
Coefficient of determination of the linear regression	0.67

Anscombe's quartet, Wikipedia

Benefits

- ❑ Facilitate understanding
 - Faster
 - Easier
- ❑ Facilitate pattern identification
 - To find outliers
 - To decide what to do next
- ❑ Attract attention
 - For communication purposes

From Data to Insights



Stephan Diehl

Definition

“A dashboard is a visual display for the most important information needed to achieve one or more objectives; consolidated and arranged on a single screen so the information can be monitored and understood at a glance.”

S. Few, 2004

A typical dashboard contains three elements:

- ❑ Diagram(s) visualizing the **metrics**
- ❑ Heading explaining the content of the dashboard and its purpose
- ❑ Short explanation of the status and information in the diagram

Dashboard characteristics

- ❑ Fits in a single computer screen
- ❑ Contains minimal information (less is more)
 - Shows the most important performance measures
- ❑ Displays data automatically
 - Updated (daily) without human assistance
- ❑ Allows filtering and drilling-down

Indicators

- Goals
 - Specific strategies
- Business Objectives
 - Desirable outcomes
 - Doable, Understandable, Manageable, Beneficial
- Metrics
 - Measurable numbers
- Key Performance Indicators
 - Metrics of objectives
- Targets
 - Pre-determined indicators of success or failure

<https://www.kaushik.net/avinash/web-analytics-101-definitions-goals-metrics-kpis-dimensions-targets>

SMART Key Performance Indicators

“A measurable value that demonstrates how effectively a company is achieving key business objectives.”

- ❑ Specific
- ❑ Measurable
- ❑ Achievable
- ❑ Relevant
- ❑ Timely

Visualization complexity

- ❑ Attributes
- ❑ Aesthetics
- ❑ Encoding
- ❑ Subset of data
- ❑ Computationally expensive
 - Approximate

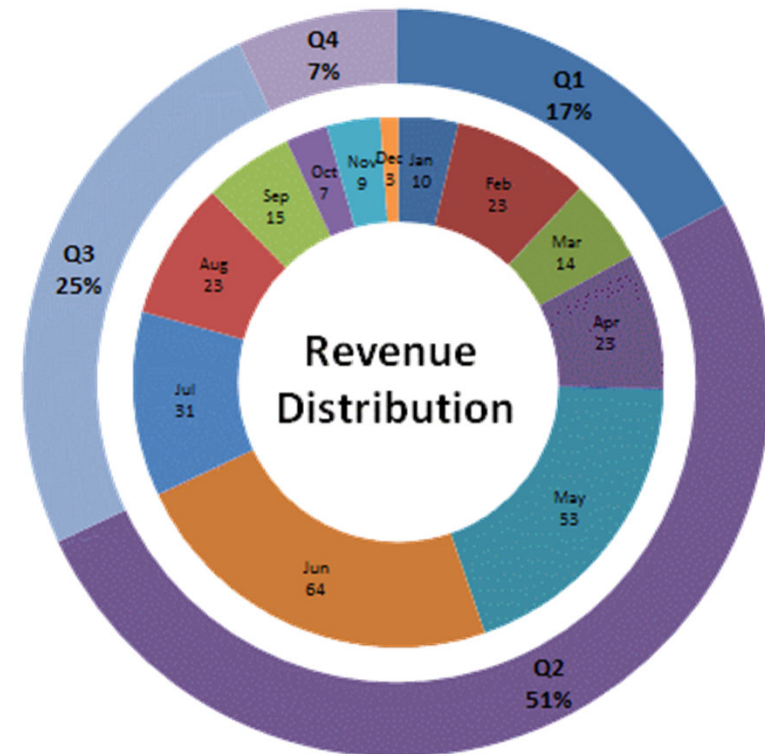
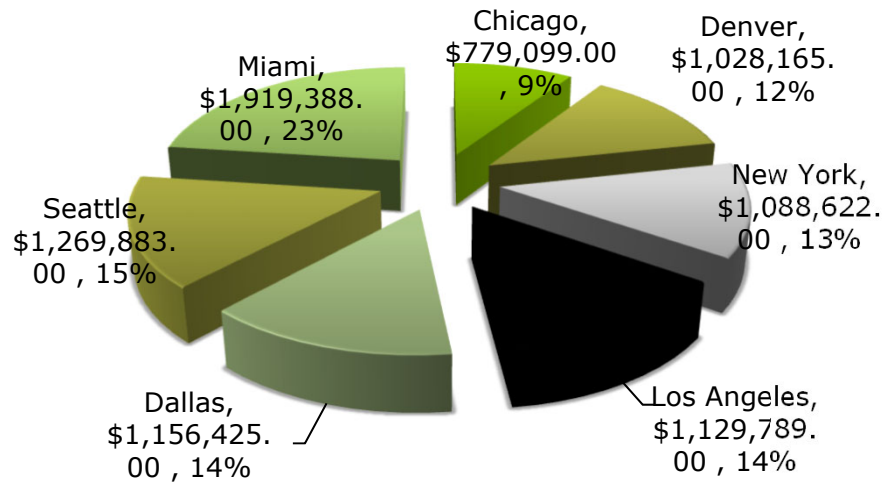
Typical problems

- ❑ Wrong kind of chart
- ❑ Confusing area and length
- ❑ Misplacement of origin
- ❑ Inconsistent scale
- ❑ Missing information
 - Title, scale, tags, etc.
- ❑ Unnecessary effects
 - Shadows, etc.
- ❑ Poor contrast between background and data
- ❑ ...

GUIDELINES

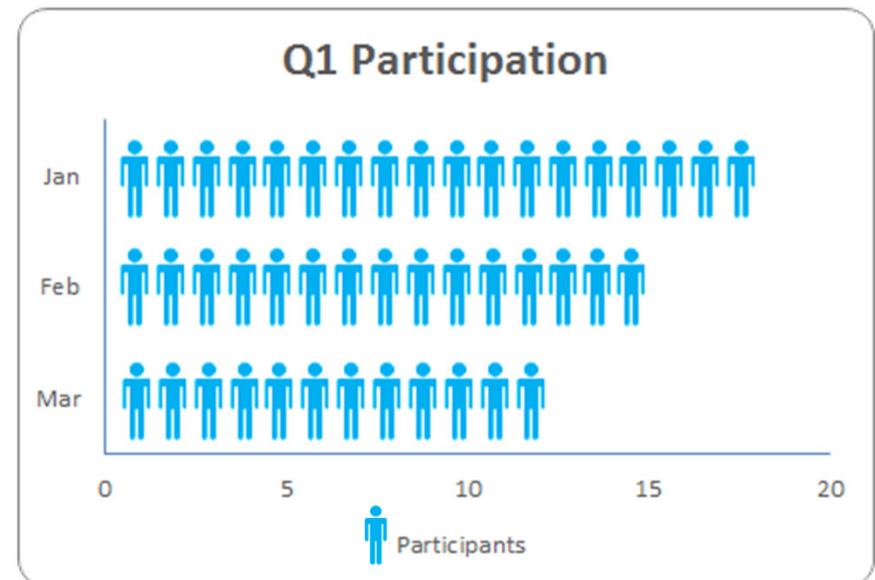
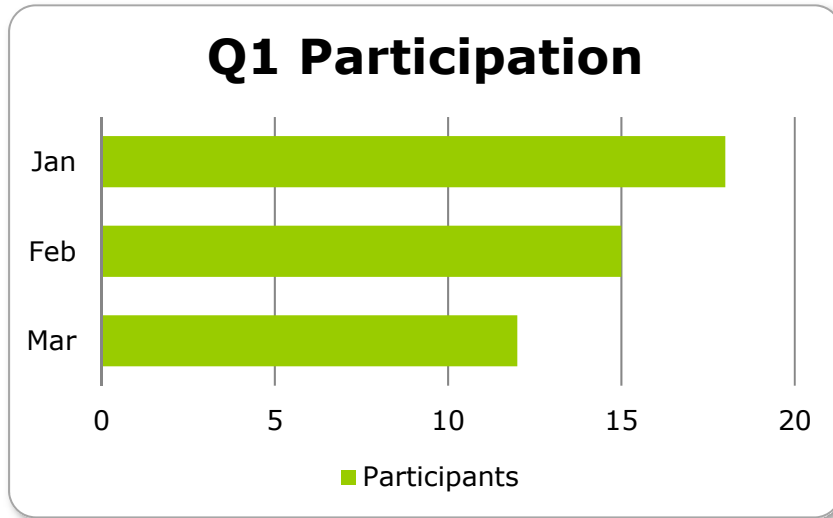
Keep it Simple ...

Earnings by City



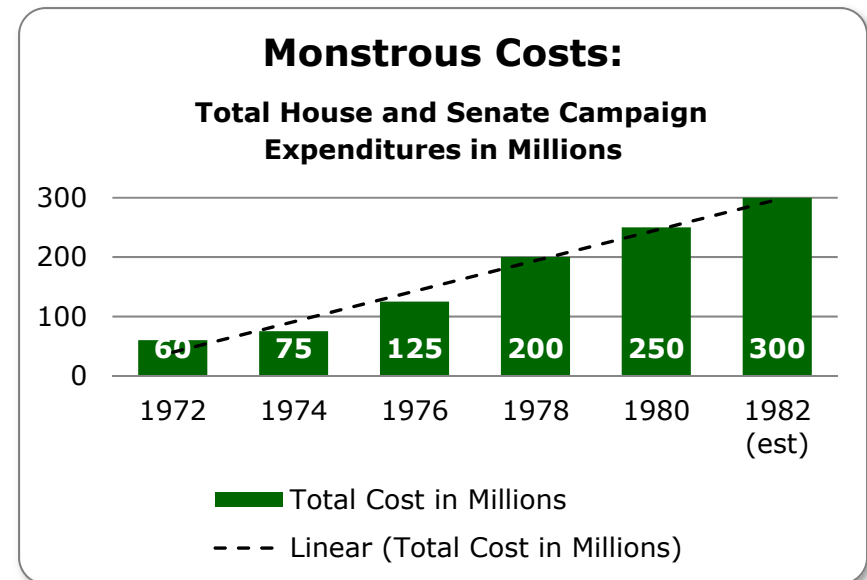
H. A. Downs

...but not unnecessarily simple



H. A. Downs

Draw the eye to your data... not your design



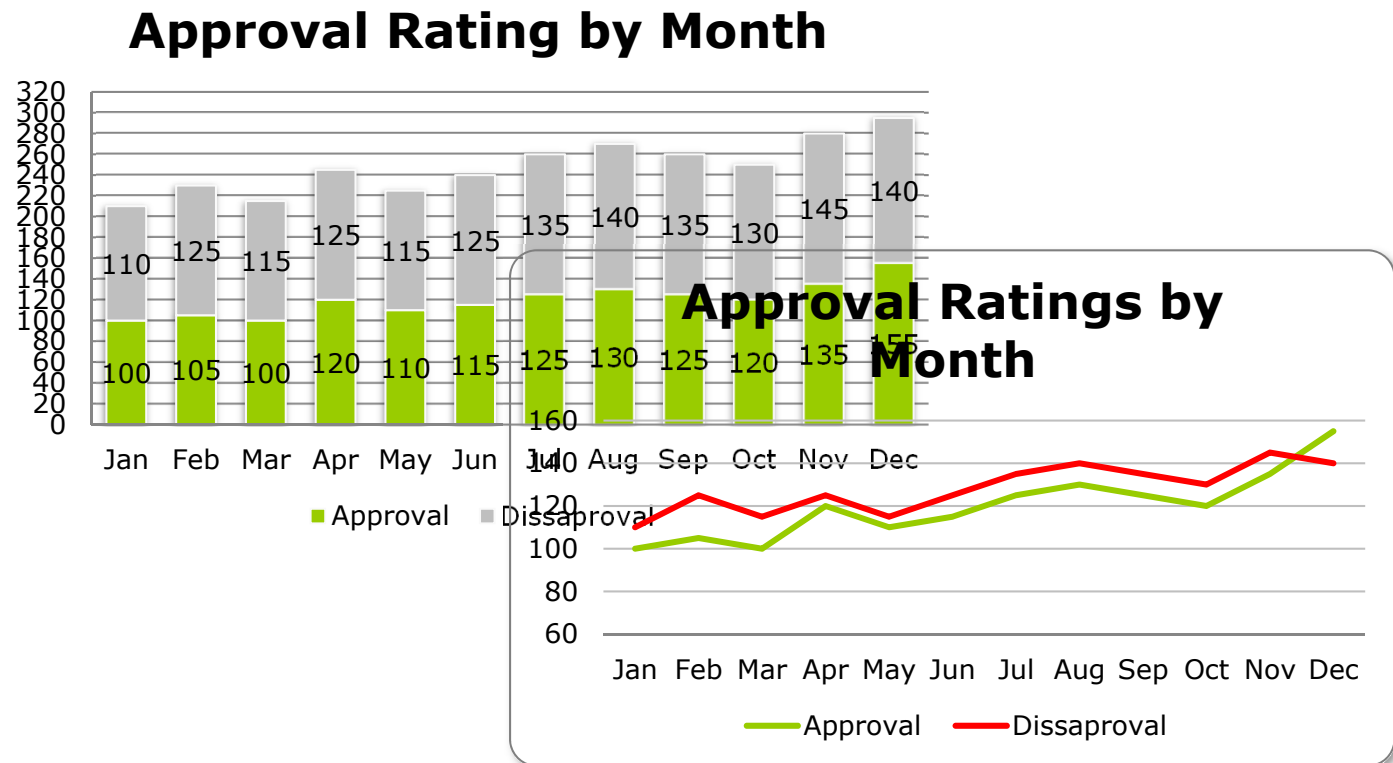
H. A. Downs

Anatomy of a Good Chart



H. A. Downs

Make it easy to make your point



H. A. Downs

DESIGN PRINCIPLES

Chunking

EDUCATION

Overview
 Sound
 Less
 Conservation Program
 Protection
 High Noise Areas
 Challenge
 Comments
 Exit

What creates sound?

the vibration of matter

Sound waves travel in all directions

Sound waves travel in all directions

Characteristics

The two main characteristics of sound are amplitude and frequency.

Amplitude is the strength of a vibration (or height of a sound wave) and is measured in decibels (dB). For every one-decibel increase, there is roughly a 20 to 30% increase in perceived loudness. The human ear can detect a human voice starting at around 5 dB, and sounds at 135 dB can cause pain. Hearing protection is recommended when you are exposed to sounds of 85 dB or greater.

Frequency is the number of sound waves in a given amount of time and is measured in hertz (Hz). One hertz equals one sound wave per second. The human ear is best adapted to hear middle-frequency sounds, about 20 to 20,000 Hz.

High-frequency sound waves make high-pitched sounds, while low-frequency sound waves make low-pitched sounds. Middle-frequency sounds in the human hearing range seem louder than sounds of higher or lower pitch.

Relating amplitude and frequency

To learn more about the relationship between amplitude and frequency, experiment with the settings below. Press the play button to see and hear the results.

How much have you learned?

Practice your knowledge by completing the activity below.

LEARNING ACTIVITY

34934137889
 +34 93 413 7889

Alignment

The image displays three different ballot layouts for the 2000 US Presidential election in Palm Beach County, Florida. Each ballot is for the general election on November 7, 2000.

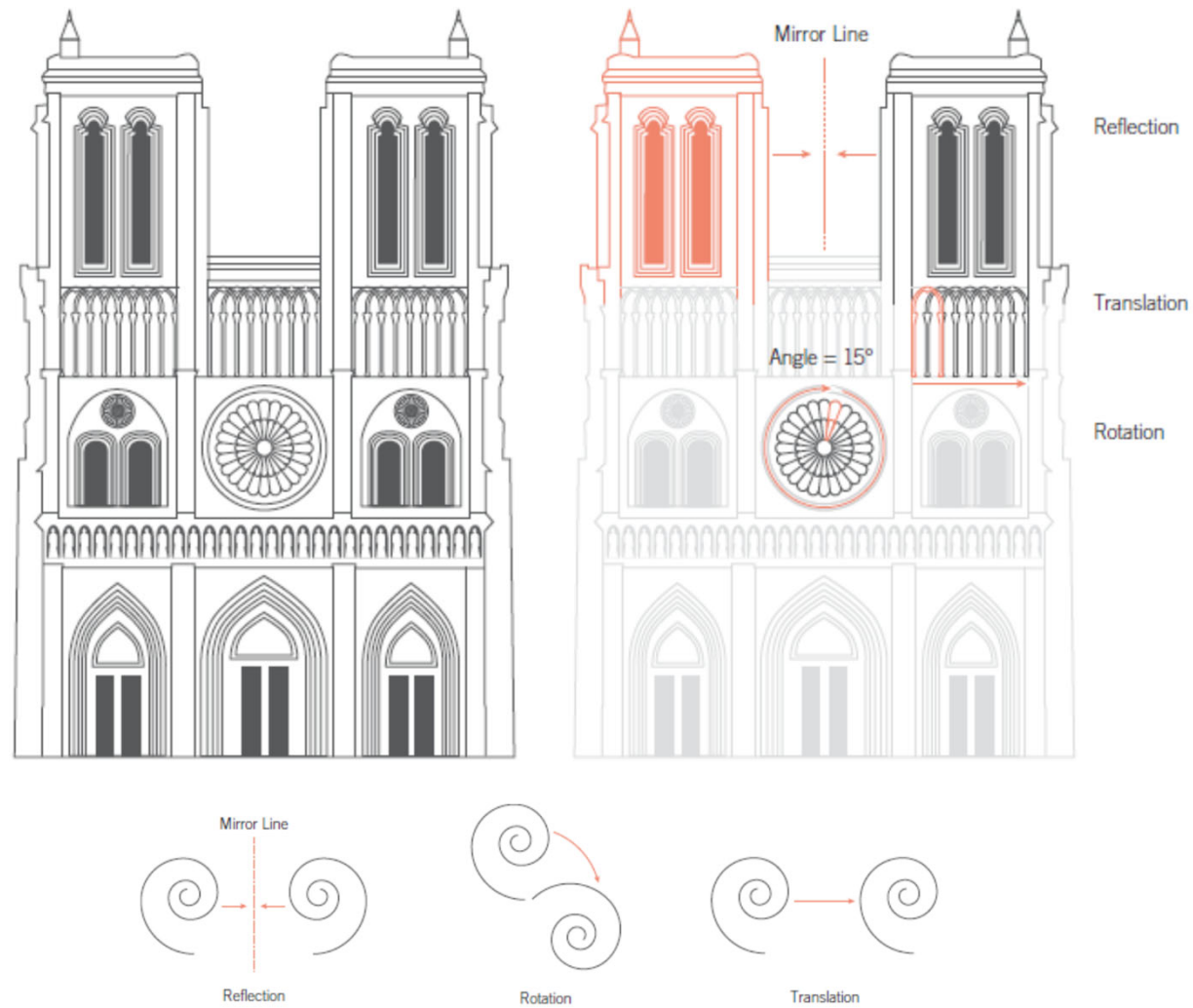
Ballot 1 (Left): Titled "OFFICIAL BALLOT, GENERAL ELECTION PALM BEACH COUNTY, FLORIDA NOVEMBER 7, 2000". It features a large box on the left for "ELECTORS FOR PRESIDENT AND VICE PRESIDENT" with a note: "(A vote for the candidates will actually be a vote for their electors.) (Vote for Group)". To the right of this box is a list of candidates grouped by party: (REPUBLICAN) GEORGE W. BUSH • PRESIDENT and DICK CHENEY • VICE PRESIDENT (3); (DEMOCRATIC) AL GORE • PRESIDENT and JOE LIEBERMAN • VICE PRESIDENT (5); (LIBERTARIAN) HARRY BROWNE • PRESIDENT and ART OLIVIER • VICE PRESIDENT (7); (GREEN) RALPH NADER • PRESIDENT and WINONA LaDUKE • VICE PRESIDENT (9); (SOCIALIST WORKERS) JAMES HARRIS • PRESIDENT and MARGARET TROWE • VICE PRESIDENT (11); (NATURAL LAW) JOHN HAGELIN • PRESIDENT and NAT GOLDHABER • VICE PRESIDENT (13). A "TURN PAGE" button is at the bottom.

Ballot 2 (Middle): Titled "OFFICIAL BALLOT, GENERAL ELECTION PALM BEACH COUNTY, FLORIDA NOVEMBER 7, 2000". It features a list of candidates grouped by party: (REFORM) PAT BUCHANAN • PRESIDENT and EZOLA FOSTER • VICE PRESIDENT (4); (SOCIALIST) DAVID McREYNOLDS • PRESIDENT and MARY CAL HOLLIS • VICE PRESIDENT (6); (CONSTITUTION) HOWARD PHILLIPS • PRESIDENT and J. CURTIS FRAZIER • VICE PRESIDENT (8); (WORKERS WORLD) MONICA MOOREHEAD • PRESIDENT and GLORIA La RIVA • VICE PRESIDENT (10). Below this is a "WRITE-IN CANDIDATE" section with instructions: "To vote for a write-in candidate follow the directions on the long stub of your ballot card." A "TURN PAGE" button is at the bottom.

Ballot 3 (Right): Titled "OFFICIAL BALLOT, GENERAL ELECTION PALM BEACH COUNTY, FLORIDA NOVEMBER 7, 2000". It features a large box on the left for "ELECTORS FOR PRESIDENT AND VICE PRESIDENT" with a note: "(A vote for the candidates will actually be a vote for their electors.) (Vote for Group)". To the right of this box is a list of candidates grouped by party: REPUBLICAN GEORGE W. BUSH • PRESIDENT AND DICK CHENEY • VICE PRESIDENT (3); DEMOCRATIC AL GORE • PRESIDENT AND JOE LIEBERMAN • VICE PRESIDENT (4); LIBERTARIAN HARRY BROWNE • PRESIDENT AND ART OLIVIER • VICE PRESIDENT (5); GREEN RALPH NADER • PRESIDENT AND WINONA LaDUKE • VICE PRESIDENT (6); SOCIALIST WORKERS JAMES HARRIS • PRESIDENT AND MARGARET TROWE • VICE PRESIDENT (7); NATURAL LAW JOHN HAGELIN • PRESIDENT AND NAT GOLDHABER • VICE PRESIDENT (8); REFORM PAT BUCHANAN • PRESIDENT AND EZOLA FOSTER • VICE PRESIDENT (9); SOCIALIST DAVID McREYNOLDS • PRESIDENT AND MARY CAL HOLLIS • VICE PRESIDENT (10); CONSTITUTION HOWARD PHILLIPS • PRESIDENT AND J. CURTIS FRAZIER • VICE PRESIDENT (11); WORKERS WORLD MONICA MOOREHEAD • PRESIDENT AND GLORIA La RIVA • VICE PRESIDENT (12). Below this is a "WRITE-IN CANDIDATE" section with instructions: "To vote for a write-in candidate, follow the directions on the long stub of your ballot card." A "TURN PAGE TO CONTINUE VOTING" button with a right-pointing arrow is at the bottom.

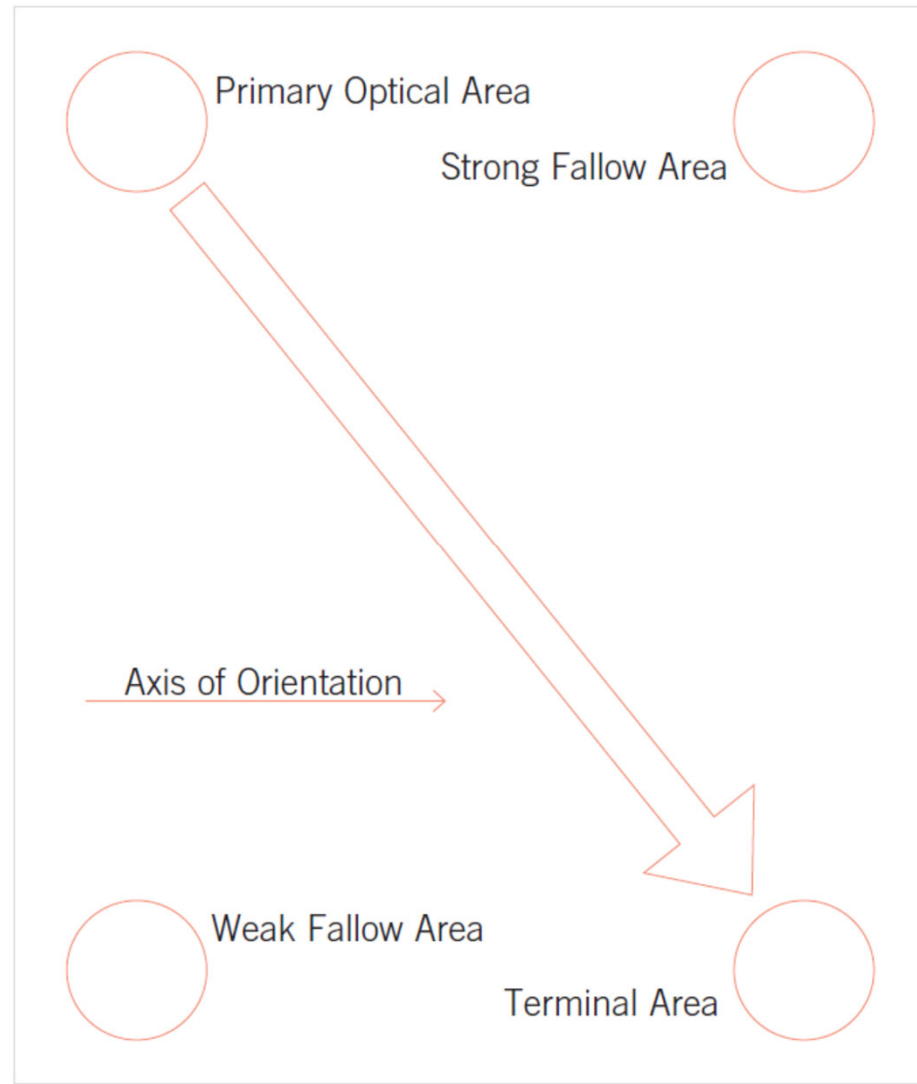
W. Lidwell et al.

Symmetry



W. Lidwell et al.

Gutenberg diagram



Highlight

General

"You mean you can't take **less**," said the Hatter, "it's **very** easy to take more than **nothing**."

"Nobody asked **your** opinion," said Alice.

"You mean you can't take **less**," said the Hatter, "it's very easy to take **more** than nothing."

"Nobody asked **your** opinion," said Alice.

Bold, Italics, and Underlining

Advice from a Caterpillar

"I can't explain **myself**, I'm afraid, sir" said Alice, "because I'm not myself, you see."

Advice from a Caterpillar

"I can't explain *myself*, I'm afraid, sir" said Alice, "because I'm not myself, you see."

Advice from a Caterpillar

"I can't explain myself, I'm afraid, sir" said Alice, "because I'm not myself, you see."

Typeface

"What is a Caucus-race?" said Alice; not that she wanted much to know, but the Dodo had paused as if it thought that somebody ought to speak, and no one else seemed inclined to say anything.

"What IS a Caucus-race?" said Alice; not that she wanted much to know, but the Dodo had paused as if it thought that **SOMEBODY** ought to speak, and no one else seemed inclined to say anything.

Color

Which brought them back again to the **beginning** of the conversation. Alice felt a **little irritated** at the Caterpillar's making such **very short** remarks, and she drew herself up and said, very gravely, "I think, you ought to tell me who **you** are, first."

Which brought them back again to the beginning of the conversation. Alice felt a little irritated at the Caterpillar's making such **very** short remarks, and she drew herself up and said, very gravely, "I think, you ought to tell me who **you** are, first."

Inversing

Who stole the tarts?



Who stole the tarts?



Colors



Analogous



Example from Nature



Triadic



Example from Nature



Complementary



Example from Nature



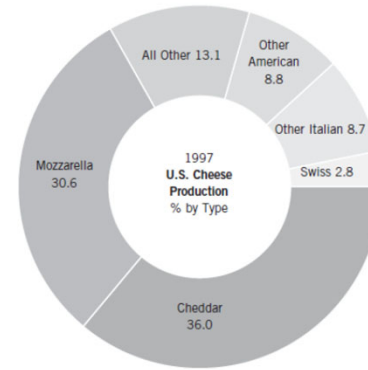
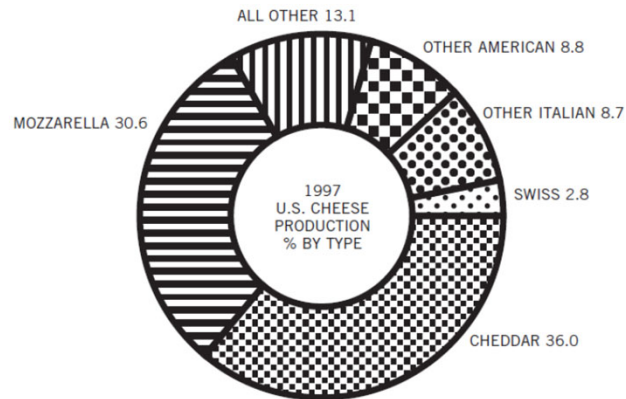
Quadratic



Example from Nature

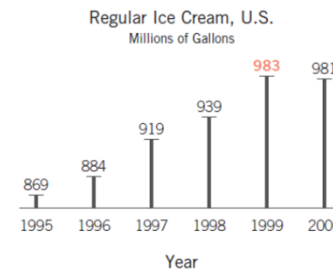
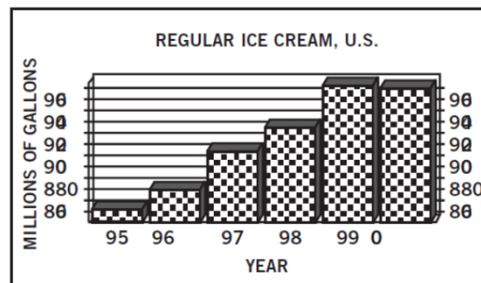
W. Lidwell et al.

Signal-to-Noise ratio

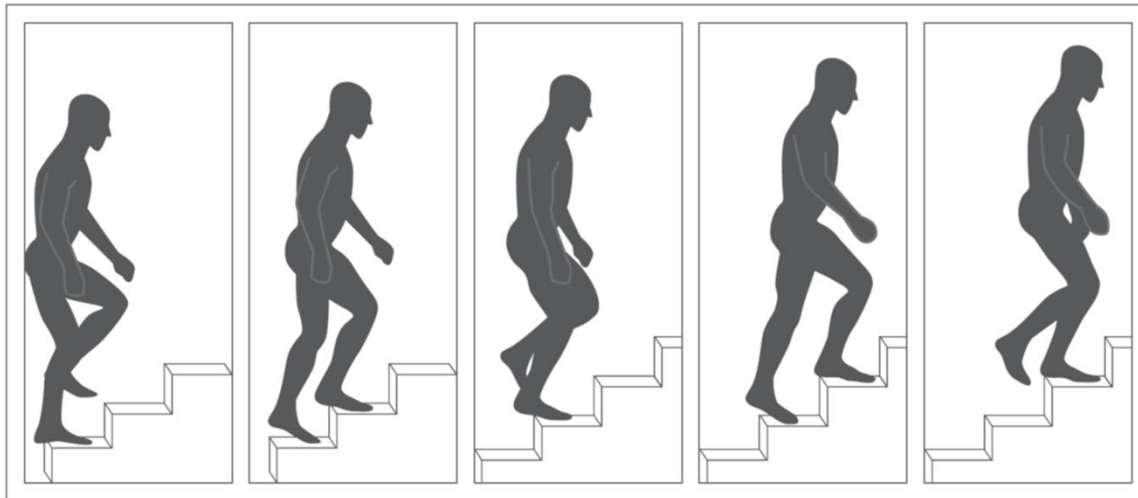


SOYBEANS		
	PRODUCTION Billions of Bushels	HARVESTED ACREAGE Millions of Acres
1997	2.69	68.1
1998	2.74	71.3
1999	2.65	72.5
2000	2.76	72.5
2001	2.89	73.2

Soybeans		
	Production Billions of Bushels	Harvested Acreage Millions of Acres
1997	2.69	68.1
1998	2.74	71.3
1999	2.65	72.5
2000	2.76	72.5
2001	2.89	73.2



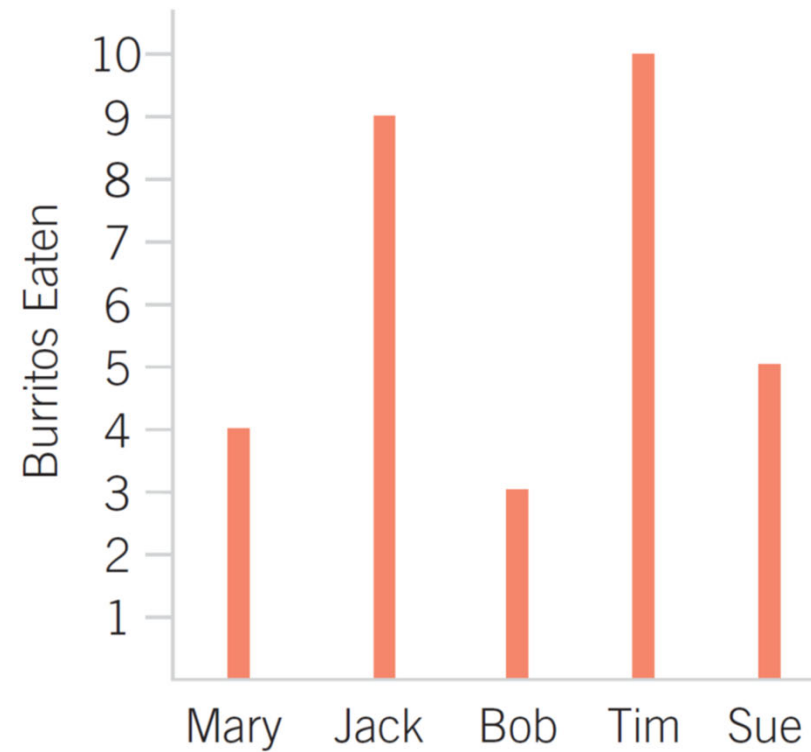
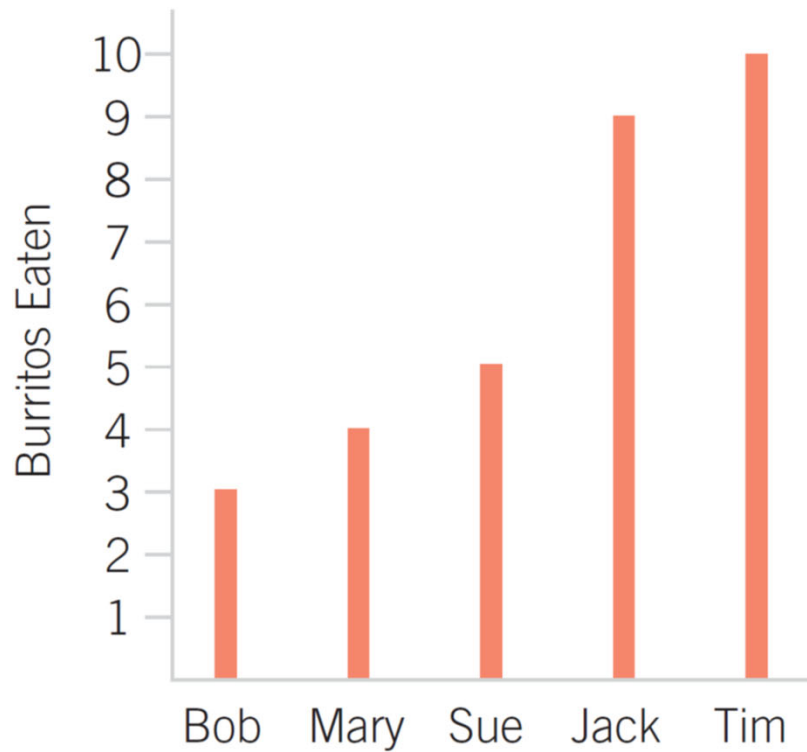
Closure



PENGUIN
ICE

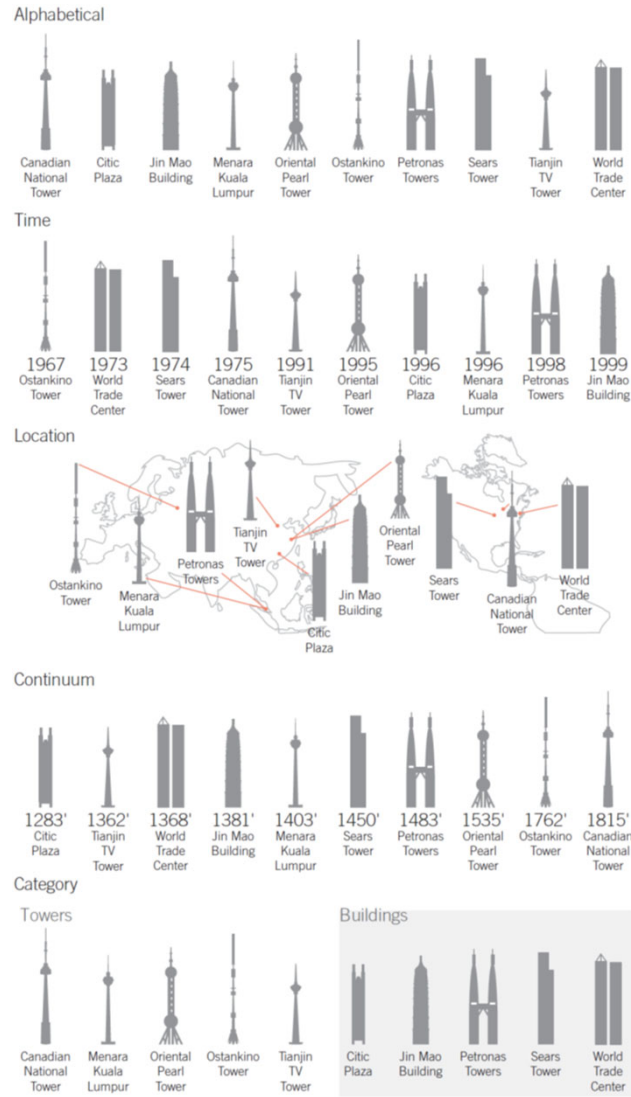
W. Lidwell et al.

Good continuation



W. Lidwell et al.

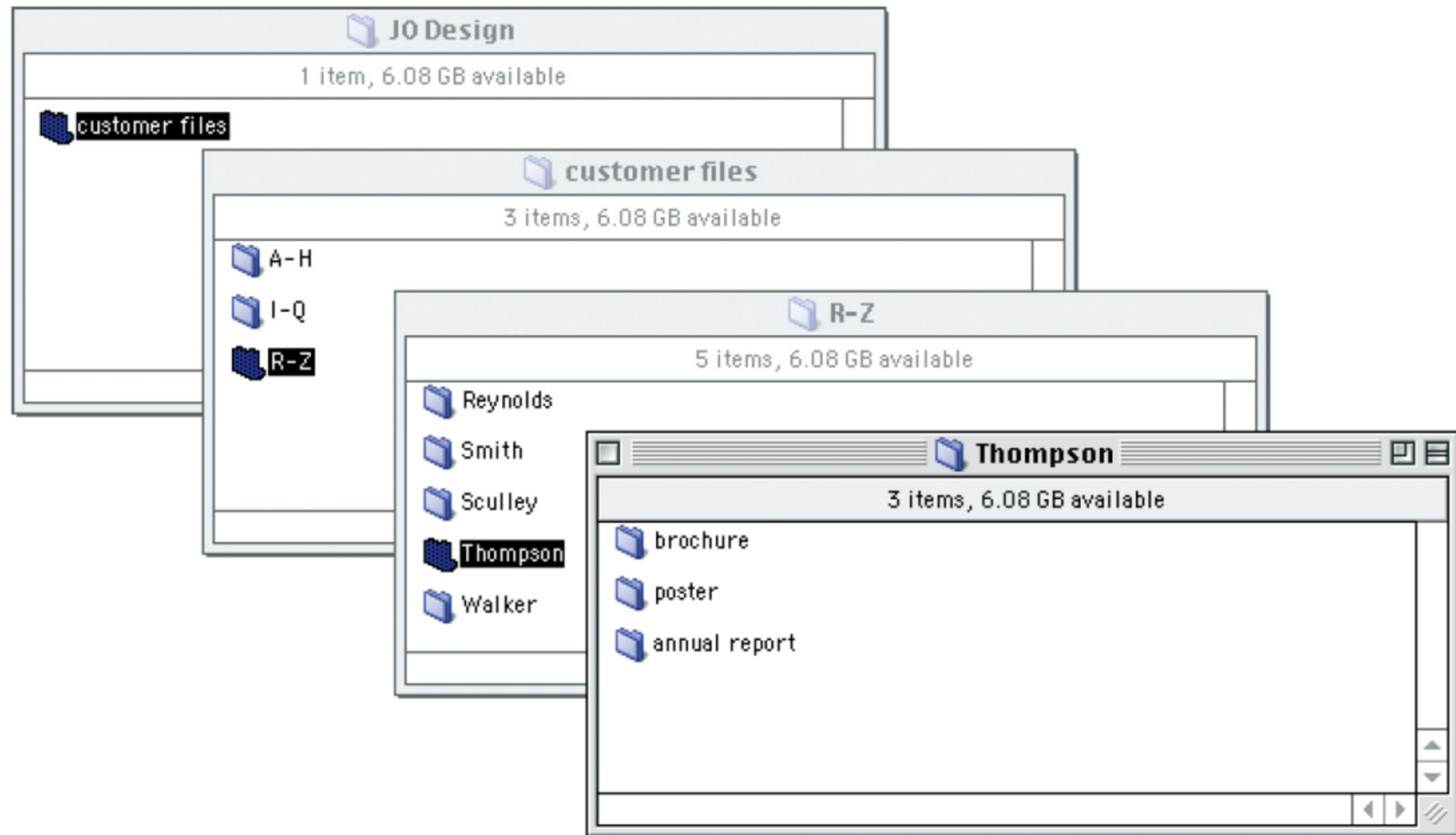
Five hat racks



Location
Alphabetical
Time
Category
Hierarchy

W. Lidwell et al.

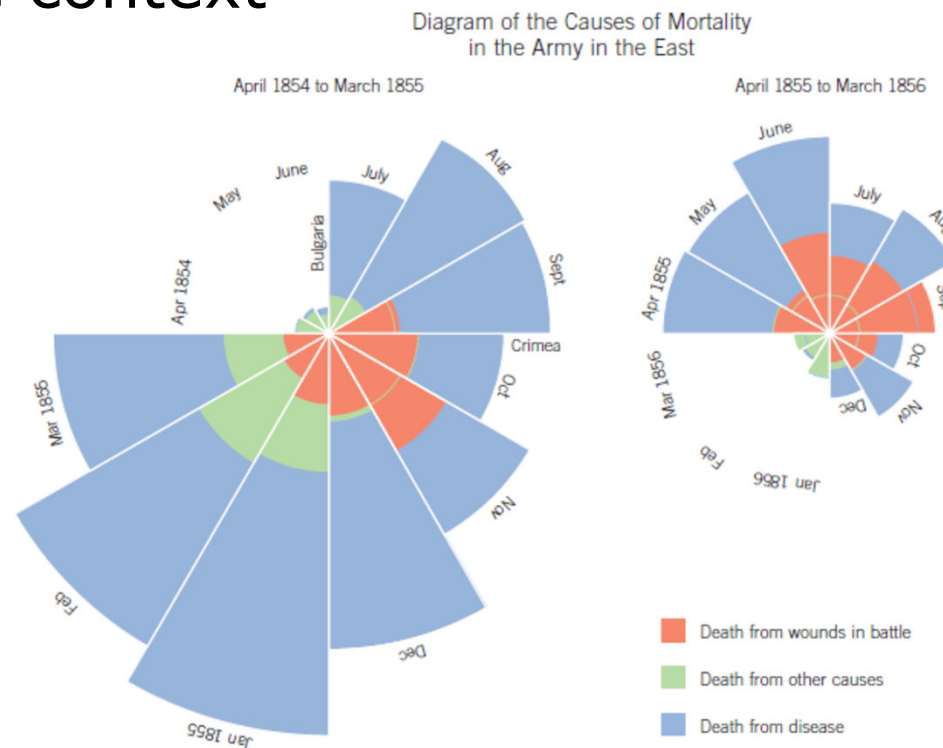
Hierarchies



W. Lidwell et al.

Comparison

- ❑ Illustrate patterns and relationships
 - Common measures and units
 - Benchmark variables for comparison
 - Single context



Layering

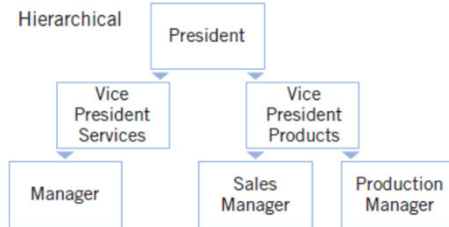
Two-Dimensional Layering

Linear

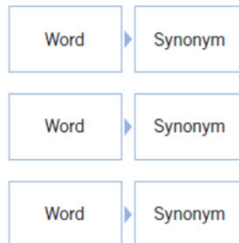
Beginning → Middle 1 → Middle 2 → End

Nonlinear

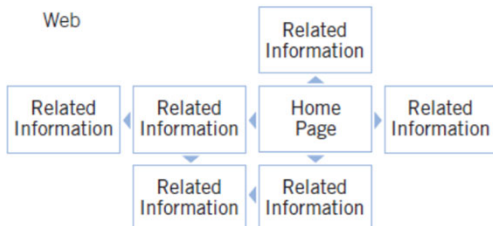
Hierarchical



Parallel



Web

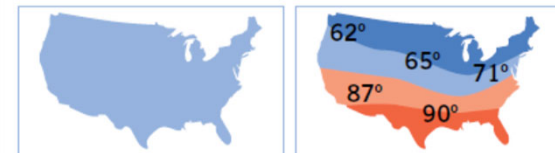


Three-Dimensional Layering

Opaque



Transparent

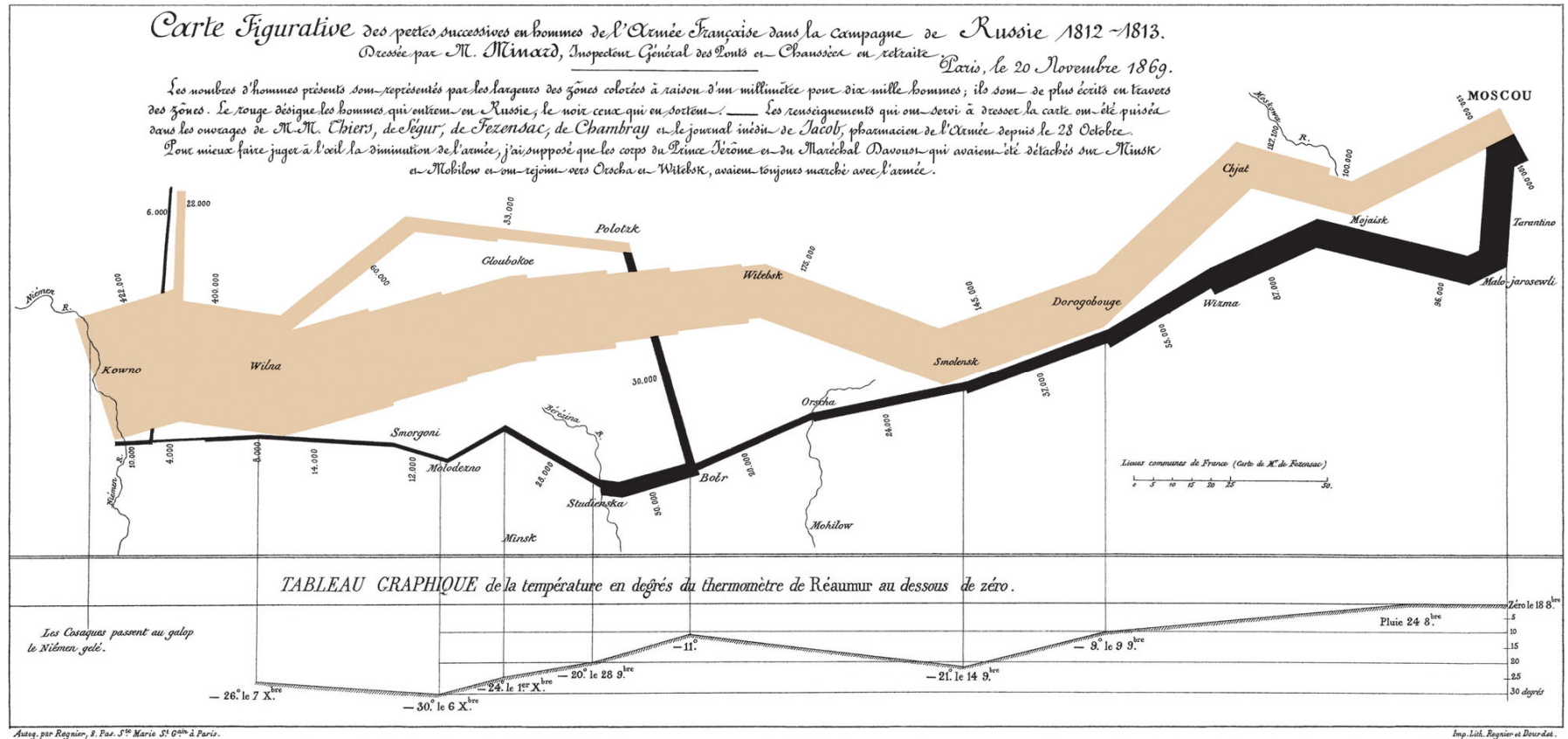


W. Lidwell et al.

Consider cognitive perception abilities

- ❑ Elements should be logically and conceptually related
- ❑ Number of elements should be no more than 7 (+2 if necessary)
 - This is the number of elements an average person can keep in the short term memory
- ❑ Use of colors should be limited to the minimum
 - They should extrapolate the diagrams and the important information in the dashboard
- ❑ Avoid 3D

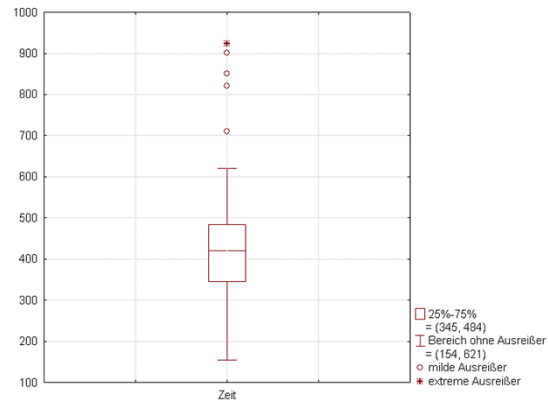
Best chart in history



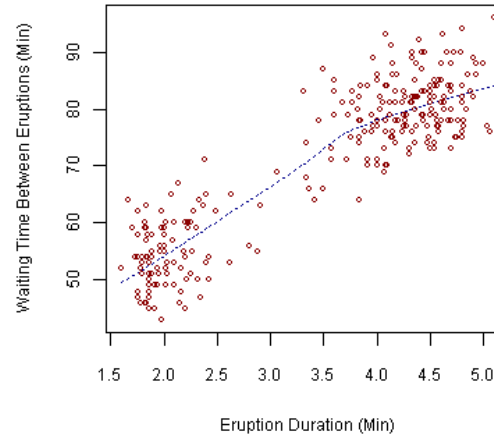
MULTIDIMENSIONAL REPRESENTATION

Representation per number of dimensions

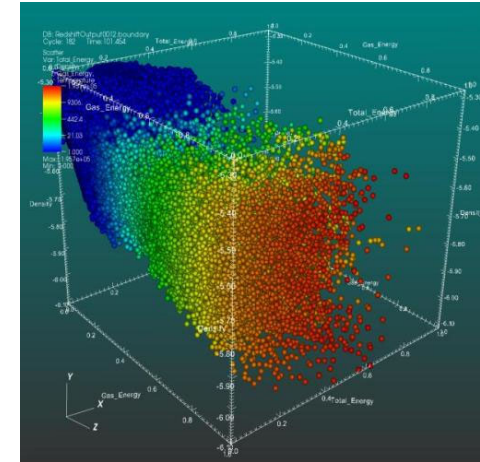
1D: Boxplot



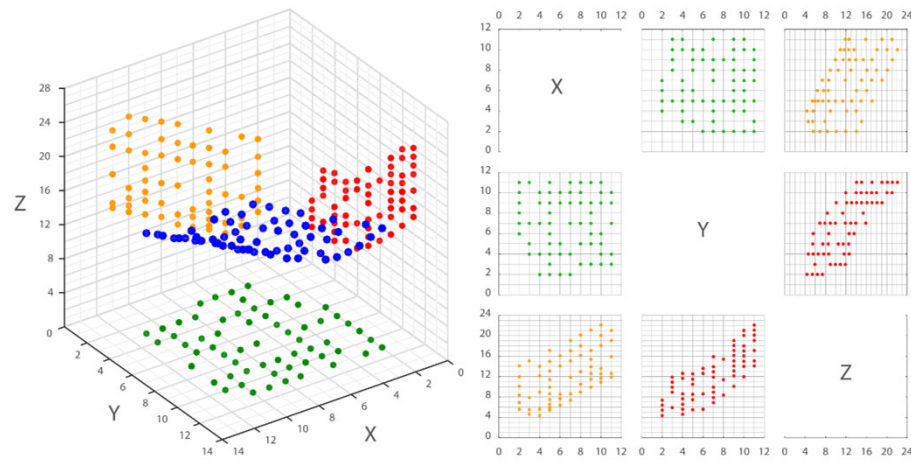
2D: Scatterplot



3D: Scatterplot

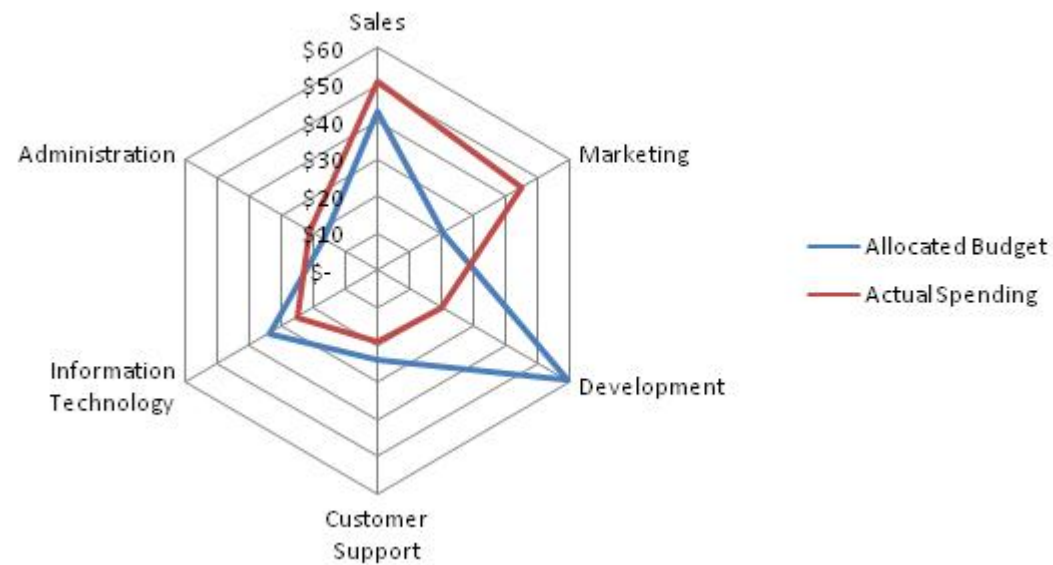


3D: Scatterplot matrix

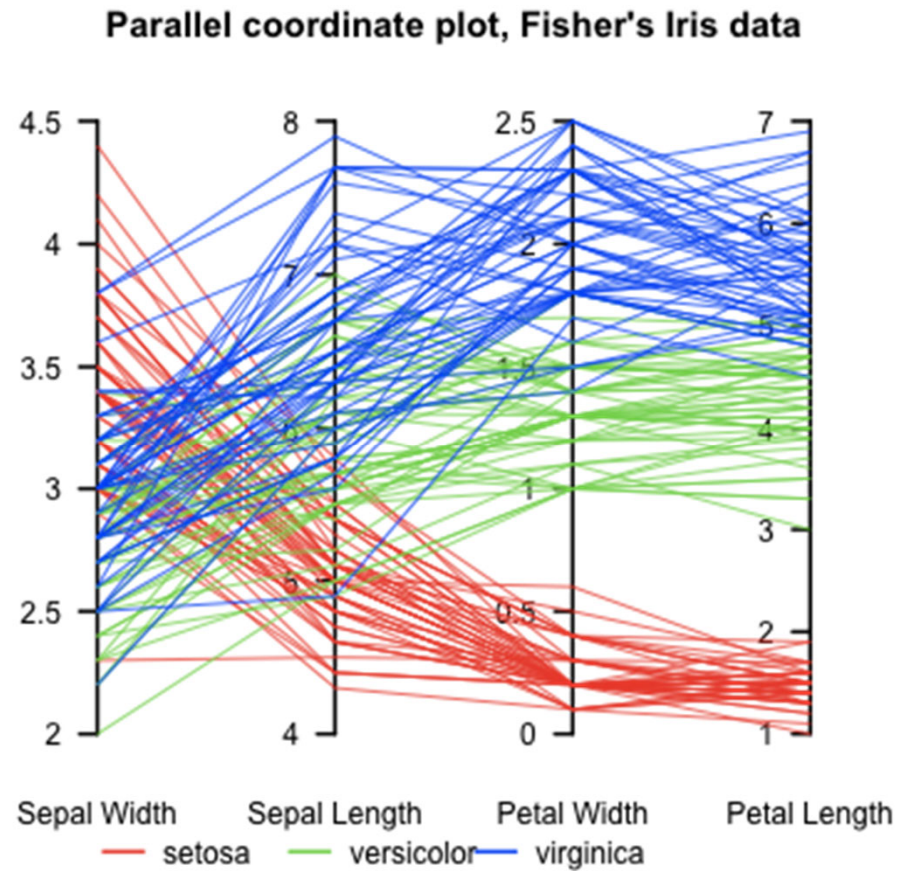


Wikipedia

Radar chart (Star plot)



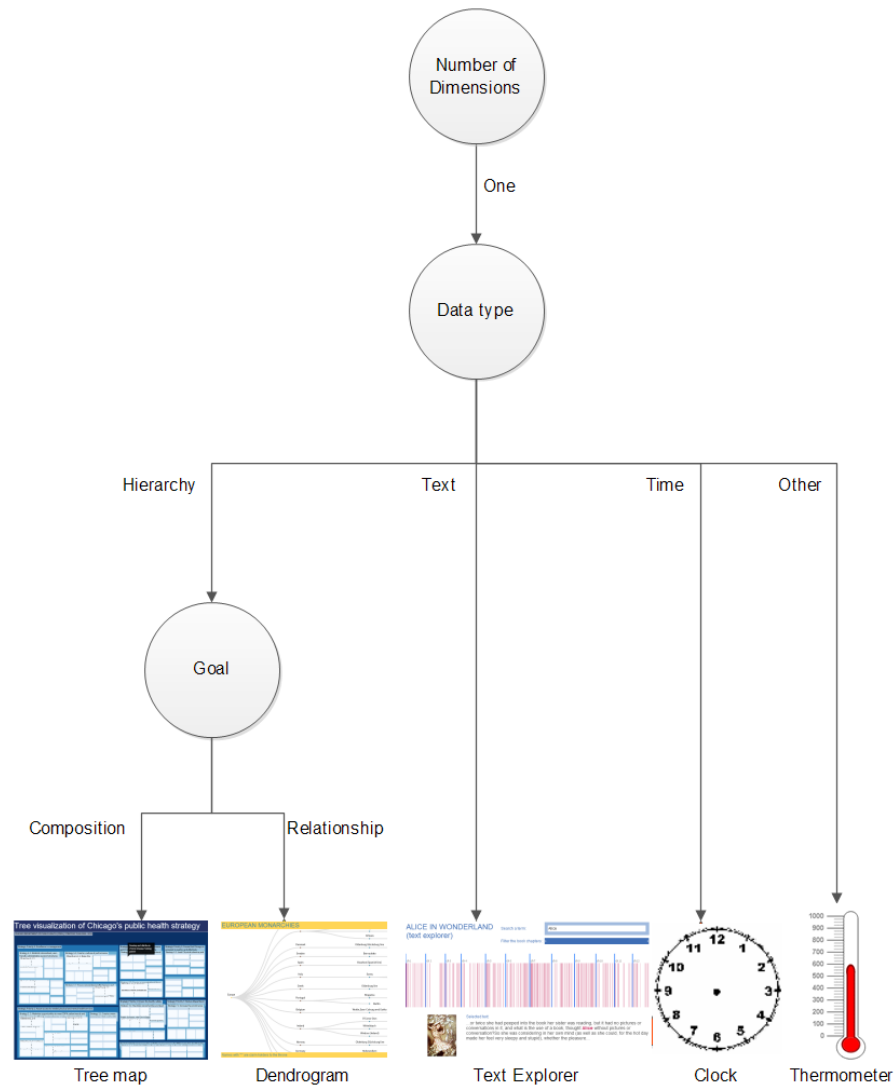
Parallel coordinates plot



Choose the right kind of chart (I)

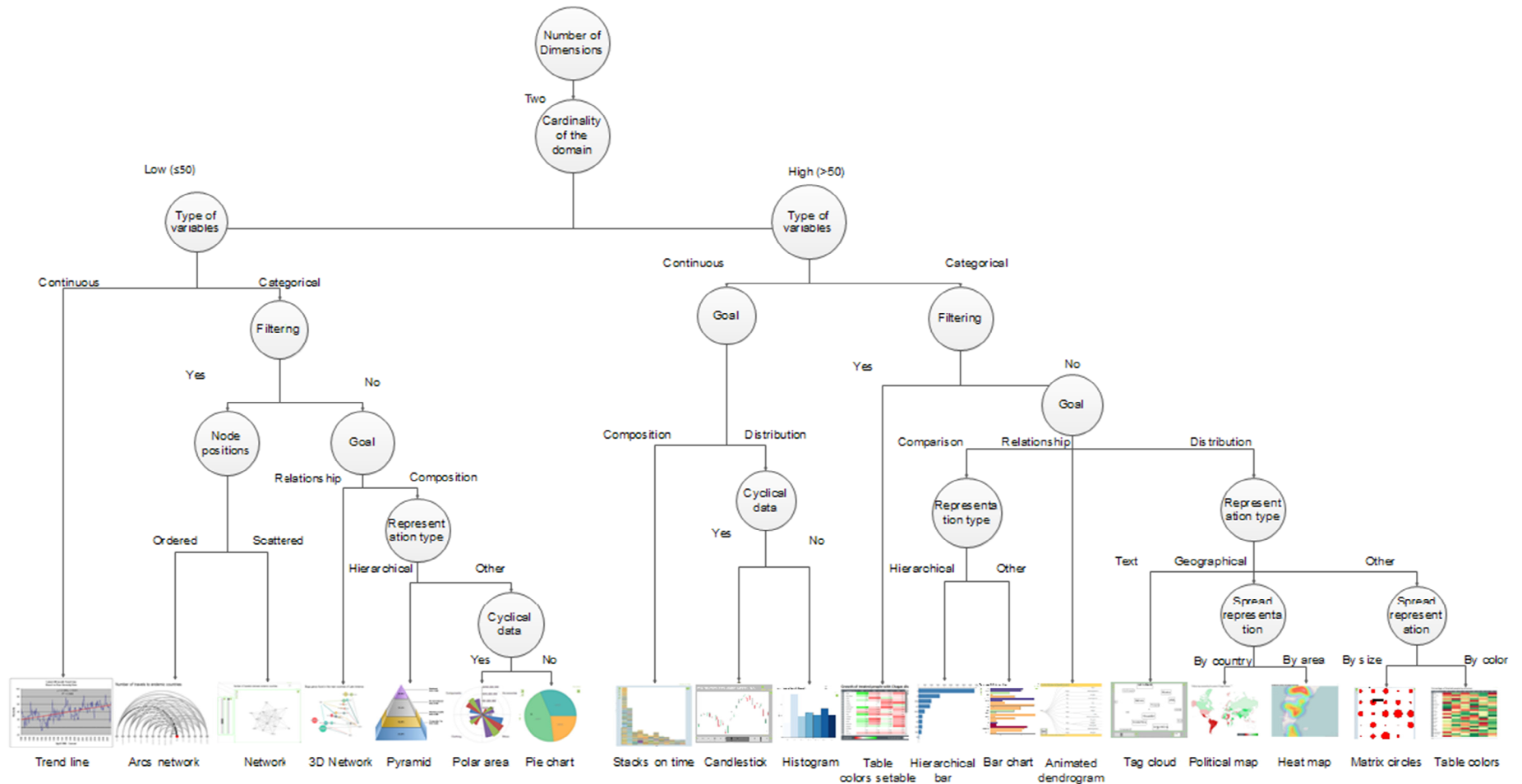
Number of dimensions	Kid of Chart	Comparison	Distribution	Relation	Proportion	Temporal evolution	Pattern	Position
1	Histogram		<u>Categorical</u>					
	Dots	<u>Categorical</u>	<u>Categorical</u>					
	Donut				<u>Continuous</u>			
	Bars				<u>Continuous</u>			
2	Time series	<u>Categorical</u>				<u>Categorical</u>	<u>Categorical</u>	
	Bivariant	<u>Categorical</u>		<u>Categorical</u>			<u>Categorical</u>	
	Bubbles	<u>Categorical</u>		<u>Categorical</u>	<u>Categorical</u>		<u>Categorical</u>	
	Mosaic			<u>Continuous</u>	<u>Continuous</u>		<u>Continuous</u>	
	Map							Both

Choose the right kind of chart (II)



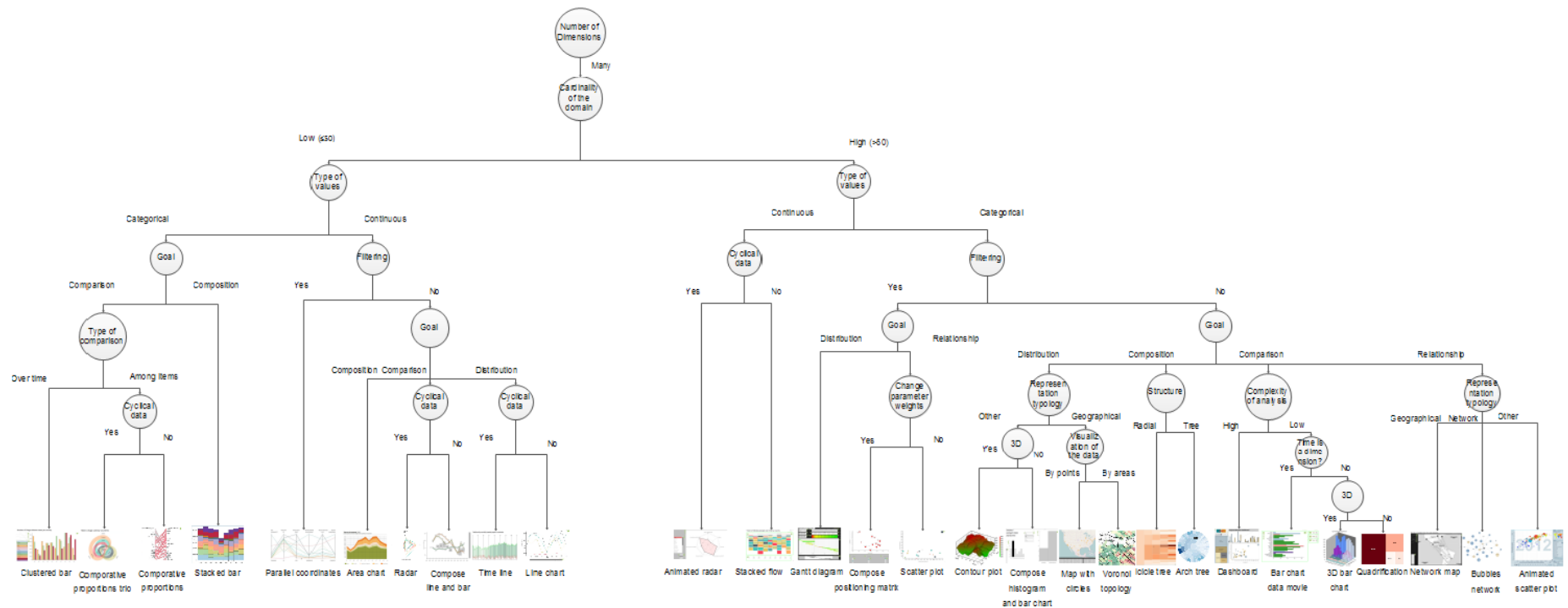
Daniele Perfetti

Choose the right kind of chart (III)



Daniele Perfetti

Choose the right kind of chart (IV)



Daniele Perfetti



"After careful consideration of all 437 charts, graphs, and metrics, I've decided to throw up my hands, hit the liquor store, and get snockered. Who's with me?!"

CLOSING

Summary

- ❑ Dashboard
 - Key Performance Indicators
- ❑ Dashboard problems
- ❑ Dashboard guidelines
- ❑ Graph design principles
- ❑ Multidimensional representation

Bibliography

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- ❑ S. K. Card et al. *Information Visualization*. Academic Press, 1999
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- ❑ H. A. Downs. *Focusing Data Dashboards – The Basics*. Evaluation 2016